As an oxidising agent: The nonmetals like carbon, sulphur, phosphorus and iodine are oxidised by nitric acid.

$$C + 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + CO_2$$

$$S + 2HNO_3 \longrightarrow H_2SO_4 + 2NO$$

$$P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 4H_2O + 20NO_2$$

$$3I_2 + 10HNO_3 \longrightarrow 6HIO_3 + 10NO + 2H_2O$$

$$HNO_3 + F_2 \longrightarrow HF + NO_3F$$

$$3H_2S + 2HNO_3 \longrightarrow 3S + 2NO + 4H_2O$$

As an nitrating agent: In organic compounds replacement of a –H atom with $-NO_2$ is often referred as nitration. For example.

$$C_6H_6 + HNO_3 \xrightarrow{H_2SO_4} C_6H_5NO_2 + H_2O$$

Nitration takes place due to the formation of nitronium ion

$$HNO_3 + H_2SO_4 \longrightarrow NO_2^+ + H_3O^+ + HSO_4^-$$

Action of nitric acid on metals

All metals with the exception of gold, platinum, rhodium, iridium and tantalum reacts with nitric acid. Nitric acid oxidises the metals. Some metals such as aluminium, iron, cobalt, nickel and chromium are rendered passive in concentrated acid due to the formation of a layer of their oxides on the metal surface, which prevents the nitric acid from reacting with pure metal.

With weak electropositive metals like tin, arsenic, antimony, tungsten and molybdenum, nitric acid gives metal oxides in which the metal is in the higher oxidation state and the acid is reduced to a lower oxidation state. The most common products evolved when nitric acid reacts with a metal are gases NO_2, NO and H_2O. Occasionally N_2, NH_2OH and NH_3 are also formed.

$$\overset{+5}{HNO_3} \quad \overset{+4}{NO_2} \quad \overset{+3}{HNO_2} \quad \overset{+2}{NO} \quad \overset{+1}{N_2O} \quad \overset{0}{N_2} \quad \overset{-3}{NH_3}$$

The reactions of metals with nitric acid are explained in 3 steps as follows:

Primary reaction: Metal nitrate is formed with the release of nascent hydrogen

$$M + HNO_3 \longrightarrow MNO_3 + (H)$$

Secondary reaction: Nascent hydrogen produces the reduction products of nitric acid.

$$HNO_3 + 2H \longrightarrow \underset{\text{Nitrous acid}}{HNO_2} + H_2O$$

$$HNO_3 + 6H \longrightarrow \underset{\text{Hydroxylamine}}{NH_2OH} + 2H_2O$$

$$HNO_3 + 8H \longrightarrow \underset{\text{Ammonia}}{NH_3} + 3H_2O$$

$$2HNO_3 + 8H \longrightarrow \underset{\text{Hypo nitrous acid}}{H_2N_2O_2} + 4H_2O$$

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